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Subject: Programming Fundamentals Labs **Section:** B

LAB # 6

Tasks

6.1: Nested 'for' Statement (Drawing Patterns with Nested for Loops)

Write a program that uses for statements to print the following patterns separately, **one below the other**. Use for loops to generate the patterns. All asterisks (*) should be printed by a single statement of the form cout << '*'; (this causes the asterisks to print side by side).

[**Hint**: The last two patterns require that each line begin with an appropriate number of blanks. **Extra credit:** Combine your code from the four separate problems into a single program that prints all four

```
(a)           (b)           (c)           (d)
*             **********    **********             *
**            *********      *********            **
***           ********        ********           ***
****          *******          *******          ****
*****         ******            ******         *****
******        *****              *****        ******
*******       ****                ****       *******
********      ***                  ***      ********
*********     **                    **     *********
**********    *                      *    **********
```

A) SOURCE CODE:

```
#include <iostream>
using namespace std;
int main ()
{
	cout << "Hunaina Yasir" << "    " << "073";
	cout << "\n";

	for (int i=0 ; i<=10 ; i++ ) {
		for (int j=0 ; j<=i ; j++) {
			cout<<"*";
		}
```

```
                    cout<<"\n";
            }
            return 0;
}
```

```
#include <iostream>
using namespace std;
int main ()
{
    cout << "Hunaina Yasir" << "    " << "073";
    cout << "\n";

    for (int i=0 ; i<=10 ; i++ ) {
        for (int j=0 ; j<=i ; j++) {
            cout<<"*";
        }
        cout<<"\n";
    }
    return 0;
}
```

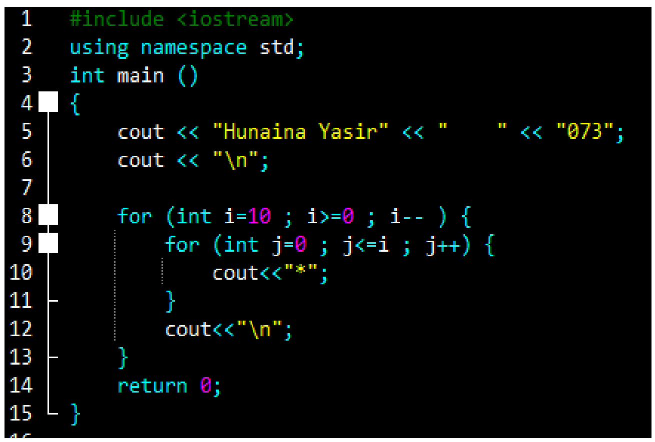
OUTPUT:

```
Hunaina Yasir    073
*
**
***
****
*****
******
*******
********
*********
**********
***********
```

```
Hunaina Yasir    073
*
**
***
****
*****
******
*******
********
*********
**********
***********
```

B) SOURCE CODE:

```
#include <iostream>
using namespace std;
int main ()
{
	cout << "Hunaina Yasir" << "    " << "073";
	cout << "\n";

	for (int i=10 ; i>=0 ; i-- ) {
		for (int j=0 ; j<=i ; j++) {
			cout<<"*";
		}
		cout<<"\n";
	}
	return 0;
}
```



```
#include <iostream>
using namespace std;
int main ()
{
    cout << "Hunaina Yasir" << "    " << "073";
    cout << "\n";

    for (int i=10 ; i>=0 ; i-- ) {
        for (int j=0 ; j<=i ; j++) {
            cout<<"*";
        }
        cout<<"\n";
    }
    return 0;
}
```

OUTPUT:

```
Hunaina Yasir    073
***********
**********
*********
********
*******
******
*****
****
***
**
```

*

```
Hunaina Yasir    073
***********
**********
*********
********
*******
******
*****
****
***
**
*
```

C) SOURCE CODE:

```
#include <iostream>
using namespace std;
int main ()
{
	cout << "Hunaina Yasir" << "    " << "073";
	cout << "\n";

	for (int i=0 ; i<=10 ; i++ )
	{
	for (int k=0 ; k<=i ; k++ )
	{

		cout<<" ";
	}
	for (int j=10 ; j>i ; j--)
	{

		cout<<"*";
	}
    cout<<"\n";
}
return 0;
}
```

```
#include <iostream>
using namespace std;
int main ()
{
      cout << "Hunaina Yasir" << "     " << "073";
      cout << "\n";

      for (int i=0 ; i<=10 ; i++ )
      {
      for (int k=0 ; k<=i ; k++ )
      {

            cout<<" ";
      }
      for (int j=10 ; j>i ; j--)
      {

            cout<<"*";
      }
      cout<<"\n";
}
return 0;
}
```

OUTPUT:

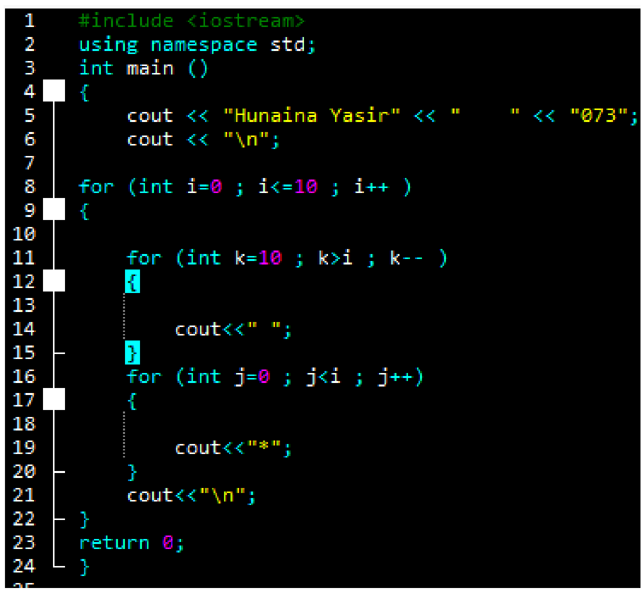
```
Hunaina Yasir    073
 **********
  *********
   ********
    *******
     ******
      *****
       ****
        ***
         **
          *
```

```
Hunaina Yasir     073
 **********
  *********
   ********
    *******
     ******
      *****
       ****
        ***
         **
          *
```

D) SOURCE CODE:

```
#include <iostream>
using namespace std;
int main ()
{
        cout << "Hunaina Yasir" << "    " << "073";
        cout << "\n";
```

```
for (int i=0 ; i<=10 ; i++ )
{

        for (int k=10 ; k>i ; k-- )
        {

                cout<<" ";
        }
        for (int j=0 ; j<i ; j++)
        {

                cout<<"*";
        }
        cout<<"\n";
}
return 0;
}
```



```
#include <iostream>
using namespace std;
int main ()
{
    cout << "Hunaina Yasir" << "     " << "073";
    cout << "\n";

for (int i=0 ; i<=10 ; i++ )
{

    for (int k=10 ; k>i ; k-- )
    {

        cout<<" ";
    }
    for (int j=0 ; j<i ; j++)
    {

        cout<<"*";
    }
    cout<<"\n";
}
return 0;
}
```

OUTPUT:

Hunaina Yasir   073

```
         *
        **
       ***
      ****
     *****
    ******
   *******
  ********
 *********
```

****

6.2: Write a C++ program to generate a 4-digit number using nested for loops and displays each number from 0000 to 9999. You will have to use 3 nested ‘for’ loops to generate the output.

SOURCE CODE:

```
#include <iostream>
using namespace std;
int main ()
{
        cout << "Hunaina Yasir" << "    " << "073";
        cout << "\n";

for (int thousands = 0; thousands <= 9; thousands++)
{
  for (int hundreds = 0; hundreds <= 9; hundreds++)
        {
    for (int tens = 0; tens <= 9; tens++)
                {
      for (int ones = 0; ones <= 9; ones++)
                        {
        cout << thousands << hundreds << tens << ones << endl;
      }
    }
  }
}
return 0;
}
```

```cpp
#include <iostream>
using namespace std;
int main ()
{
    cout << "Hunaina Yasir" << "     " << "073";
    cout << "\n";

for (int thousands = 0; thousands <= 9; thousands++)
{
    for (int hundreds = 0; hundreds <= 9; hundreds++)
    {
        for (int tens = 0; tens <= 9; tens++)
        {
            for (int ones = 0; ones <= 9; ones++)
            {
                cout << thousands << hundreds << tens << ones << endl;
            }
        }
    }
}
return 0;
}
```

OUTPUT:

```
0983
0984
0985
0986
0987
0988
0989
0990
0991
0992
0993
0994
0995
0996
0997
0998
0999
```

```
9981
9982
9983
9984
9985
9986
9987
9988
9989
9990
9991
9992
9993
9994
9995
9996
9997
9998
9999
```

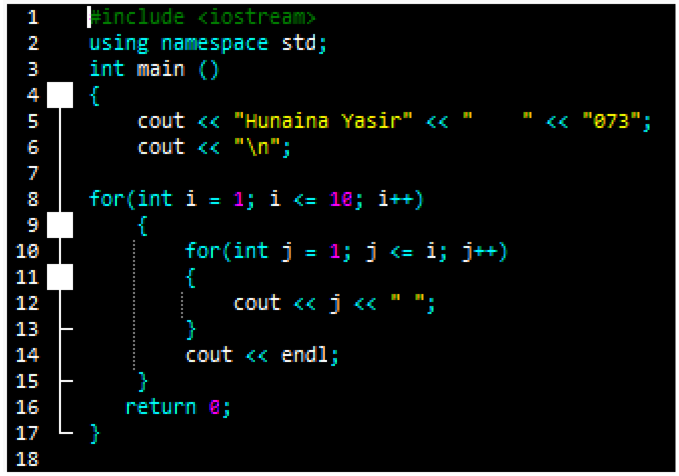
It prints from numbers 0983 to 9999.

Task 6.3: Write a C++ program to display a triangular pattern of numbers using a nested 'for' loop.

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4 5 6
1 2 3 4 5 6 7
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8 9
1 2 3 4 5 6 7 8 9 10

SOURCE CODE:

```
#include <iostream>
using namespace std;
int main ()
{
            cout << "Hunaina Yasir" << "    " << "073";
            cout << "\n";

for(int i = 1; i <= 10; i++)
   {
       for(int j = 1; j <= i; j++)
       {
          cout << j << " ";
       }
       cout << endl;
   }
  return 0;
}
```



```
#include <iostream>
using namespace std;
int main ()
{
    cout << "Hunaina Yasir" << "    " << "073";
    cout << "\n";

for(int i = 1; i <= 10; i++)
    {
        for(int j = 1; j <= i; j++)
        {
            cout << j << " ";
        }
        cout << endl;
    }
   return 0;
}
```

OUTPUT:

```
Hunaina Yasir    073
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4 5 6
1 2 3 4 5 6 7
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8 9
1 2 3 4 5 6 7 8 9 10
```

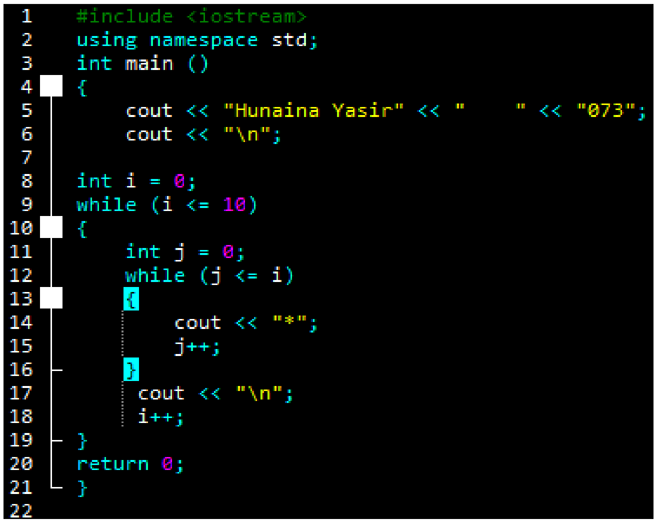
```
Hunaina Yasir    073
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4 5 6
1 2 3 4 5 6 7
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8 9
1 2 3 4 5 6 7 8 9 10
```

6.4: Repeat Task 6.1 using a nested **'while' loop**.

A) SOURCE CODE:

```
#include <iostream>
using namespace std;
int main ()
{
            cout << "Hunaina Yasir" << "    " << "073";
            cout << "\n";

int i = 0;
while (i <= 10)
{
   int j = 0;
   while (j <= i)
            {
```

```
        cout << "*";
        j++;
    }
    cout << "\n";
    i++;
}
return 0;
}
```



OUTPUT:

```
Hunaina Yasir    073
*
**
***
****
*****
******
*******
********
*********
**********
***********
```

B) SOURCE CODE:

```
#include <iostream>
using namespace std;
int main ()
{
        cout << "Hunaina Yasir" << "    " << "073";
        cout << "\n";

int i = 10;
```

```
 while (i >= 0)
 {
    int j = 0;
    while (j <= i)
           {
       cout << "*";
       j++;
    }
    cout << "\n";
    i--;
 }
return 0;
}
```

```
#include <iostream>
using namespace std;
int main ()
{
    cout << "Hunaina Yasir" << "     " << "073";
    cout << "\n";

int i = 10;
  while (i >= 0)
  {
      int j = 0;
      while (j <= i)
      {
          cout << "*";
          j++;
      }
      cout << "\n";
      i--;
  }
return 0;
}
```

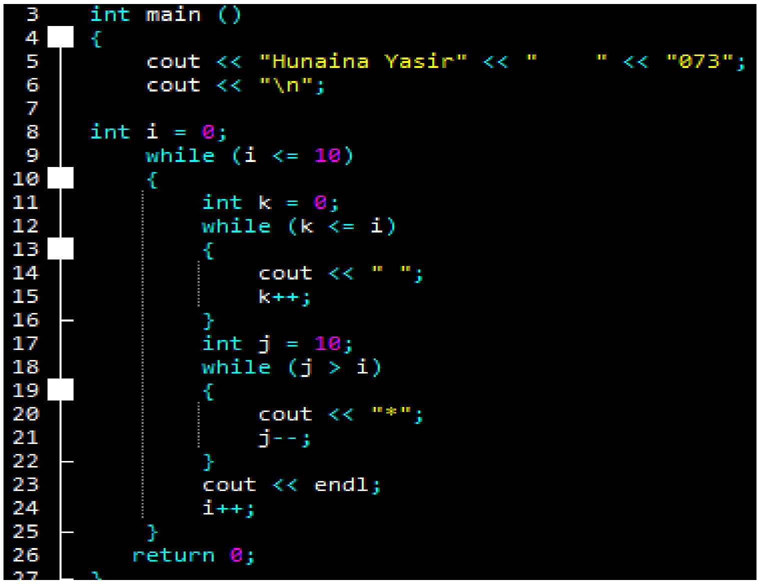
OUTPUT:

```
Hunaina Yasir    073
***********
**********
*********
********
*******
******
*****
****
***
**
*
```

```
Hunaina Yasir    073
***********
**********
*********
********
*******
******
*****
****
***
**
*
```

C) SOURCE CODE:

```
#include <iostream>
using namespace std;
int main ()
{
        cout << "Hunaina Yasir" << "    " << "073";
        cout << "\n";

int i = 0;
  while (i <= 10)
        {
    int k = 0;
    while (k <= i)
              {
      cout << " ";
      k++;
    }
    int j = 10;
    while (j > i)
              {
      cout << "*";
      j--;
    }
    cout << endl;
    i++;
  }
  return 0;
}
```



```cpp
int main ()
{
    cout << "Hunaina Yasir" << "     " << "073";
    cout << "\n";

int i = 0;
    while (i <= 10)
    {
        int k = 0;
        while (k <= i)
        {
            cout << " ";
            k++;
        }
        int j = 10;
        while (j > i)
        {
            cout << "*";
            j--;
        }
        cout << endl;
        i++;
    }
   return 0;
}
```

OUTPUT:

```
Hunaina Yasir    073
 **********
  *********
   ********
    *******
     ******
      *****
       ****
        ***
         **
          *
```

```
Hunaina Yasir    073
 **********
  *********
   ********
    *******
     ******
      *****
       ****
        ***
         **
          *
```

D) SOURCE CODE:

```cpp
#include <iostream>
using namespace std;
int main ()
{
        cout << "Hunaina Yasir" << "     " << "073";
        cout << "\n";
```

```
int i = 0;
   while (i <= 10) {
      int k = 10;
      while (k > i)
                     {
         cout << " ";
         k--;
      }
      int j = 0;
      while (j < i)
                     {
         cout << "*";
         j++;
      }
      cout << "\n";
      i++;
   }
   return 0;
}
```

```
using namespace std;
int main ()
{
     cout << "Hunaina Yasir" << "     " << "073";
     cout << "\n";

int i = 0;
     while (i <= 10) {
          int k = 10;
          while (k > i)
          {
               cout << " ";
               k--;
          }
          int j = 0;
          while (j < i)
          {
               cout << "*";
               j++;
          }
          cout << "\n";
          i++;
     }
     return 0;
}
```

OUTPUT:

Hunaina Yasir   073

```
         *
        **
       ***
      ****
     *****
    ******
   *******
  ********
```

```
 *********
**********
```

```
Hunaina Yasir    073

         *
        **
       ***
      ****
     *****
    ******
   *******
  ********
 *********
**********
```

<u>6.5:</u> Repeat Task 6.2 using nested **‘while’ loop.**

<u>SOURCE CODE:</u>

```
#include <iostream>
using namespace std;
int main ()
{
        cout << "Hunaina Yasir" << "    " << "073";
        cout << "\n";

int thousands = 0;
  while (thousands <= 9)
        {
    int hundreds = 0;
    while (hundreds <= 9)
                {
      int tens = 0;
      while (tens <= 9)
                        {
        int ones = 0;
        while (ones <= 9)
                                {
          cout << thousands << hundreds << tens << ones << endl;
          ones++;
        }

        tens++;
      }

      hundreds++;
    }

    thousands++;
  }
```

return 0;
}

```cpp
int thousands = 0;
    while (thousands <= 9)
    {
        int hundreds = 0;
        while (hundreds <= 9)
        {
            int tens = 0;
            while (tens <= 9)
            {
                int ones = 0;
                while (ones <= 9)
                {
                    cout << thousands << hundreds << tens << ones << endl;
                    ones++;
                }

                tens++;
            }

            hundreds++;
        }

        thousands++;
    }
return 0;
}
```

<u>OUTPUT:</u>

```
0983
0984
0985
0986
0987
0988
0989
0990
0991
0992
0993
0994
0995
0996
0997
0998
0999
```

```
9981
9982
9983
9984
9985
9986
9987
9988
9989
9990
9991
9992
9993
9994
9995
9996
9997
9998
9999
```

It prints from numbers 0983 to 9999.

<u>Task 6.6:</u> Write a C++ program to display a triangular pattern of numbers using a nested 'while' loop.

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

1 2 3 4 5 6
1 2 3 4 5 6 7
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8 9
1 2 3 4 5 6 7 8 9 10

SOURCE CODE:

```
#include <iostream>
using namespace std;
int main ()
{
	cout << "Hunaina Yasir" << "    " << "073";
	cout << "\n";

int i = 1;

   while (i <= 10)
	{
      int j = 1;

      while (j <= i)
		{
         cout << j << " ";
         j++;
      }

      cout << endl;
      i++;
   }

   return 0;
}
```

```cpp
#include <iostream>
using namespace std;
int main ()
{
    cout << "Hunaina Yasir" << "    " << "073";
    cout << "\n";

int i = 1;

    while (i <= 10)
    {
        int j = 1;

        while (j <= i)
        {
            cout << j << " ";
            j++;
        }

        cout << endl;
        i++;
    }

    return 0;
}
```

<u>OUTPUT:</u>

Hunaina Yasir   073
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4 5 6
1 2 3 4 5 6 7
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8 9
1 2 3 4 5 6 7 8 9 10

```
Hunaina Yasir    073
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4 5 6
1 2 3 4 5 6 7
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8 9
1 2 3 4 5 6 7 8 9 10
```

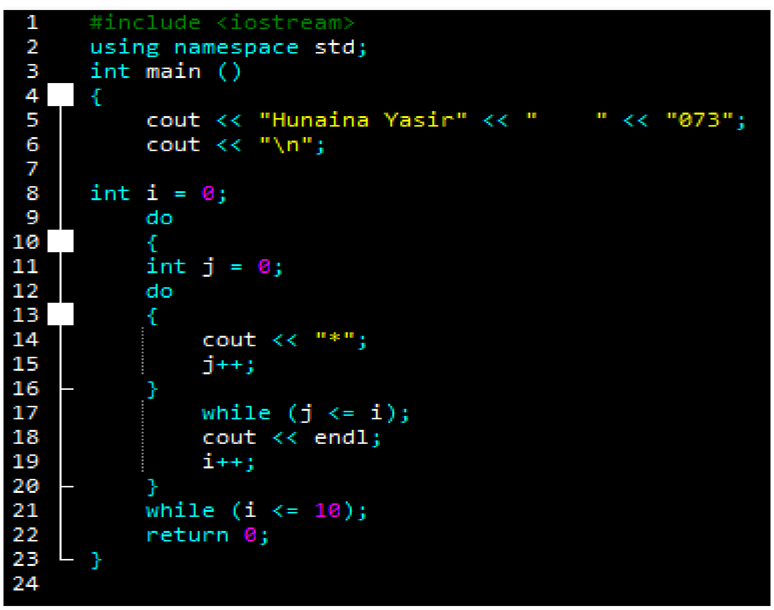
<u>Task 6.7</u> Write a C++ program to display a triangular pattern using a nested ‘do-while’ loop.

```
*
**
***
****
*****
******
*******
********
*********
**********
```

```cpp
int i = 0;
    do {
        int j = 0;
        do {
            cout << "*";
            j++;
        } while (j <= i);
        cout << endl;
        i++;
    } while (i <= 10);
```

<u>SOURCE CODE:</u>

```
#include <iostream>
using namespace std;
int main ()
{
	cout << "Hunaina Yasir" << "    " << "073";
	cout << "\n";

int i = 0;
   do
	{
   int j = 0;
   do
	{
      cout << "*";
      j++;
   }
		while (j <= i);
      cout << endl;
      i++;
   }
	while (i <= 10);
	return 0;
}
```



```
#include <iostream>
using namespace std;
int main ()
{
	cout << "Hunaina Yasir" << "     " << "073";
	cout << "\n";

int i = 0;
	do
	{
	int j = 0;
	do
	{
		cout << "*";
		j++;
	}
		while (j <= i);
		cout << endl;
		i++;
	}
	while (i <= 10);
	return 0;
}
```

OUTPUT:

Hunaina Yasir   073
*
**
***
****
*****

```
******
*******
********
*********
**********
***********
```

<u>6.8:</u> Repeat Task 6.2 using nested **‘do-while’ loop.**

<u>SOURCE CODE:</u>

```
#include <iostream>
using namespace std;
int main ()
{
        cout << "Hunaina Yasir" << "    " << "073";
        cout << "\n";

int thousands = 0;
   do {
      int hundreds = 0;
      do {
         int tens = 0;
         do {
            int ones = 0;
            do {
               cout << thousands << hundreds << tens << ones << endl;
               ones++;
            } while (ones <= 9);
            tens++;
         } while (tens <= 9);
         hundreds++;
      } while (hundreds <= 9);
      thousands++;
   } while (thousands <= 9);
```

```
return 0;
}
```

```
int main ()
{
    cout << "Hunaina Yasir" << "     " << "073";
    cout << "\n";

int thousands = 0;
    do {
        int hundreds = 0;
        do {
            int tens = 0;
            do {
                int ones = 0;
                do {
                    cout << thousands << hundreds << tens << ones << endl;
                    ones++;
                } while (ones <= 9);
                tens++;
            } while (tens <= 9);
            hundreds++;
        } while (hundreds <= 9);
        thousands++;
    } while (thousands <= 9);

    return 0;
}
```

OUTPUT:

```
0983
0984
0985
0986
0987
0988
0989
0990
0991
0992
0993
0994
0995
0996
0997
0998
0999
```

```
9981
9982
9983
9984
9985
9986
9987
9988
9989
9990
9991
9992
9993
9994
9995
9996
9997
9998
9999
```

It prints from numbers 0983 to 9999.

6.9: Write a C++ program to display a triangular pattern of numbers using a nested ‘do-while’ loop.

1

1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4 5 6
1 2 3 4 5 6 7
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8 9
1 2 3 4 5 6 7 8 9 10

SOURCE CODE:

```
#include <iostream>
using namespace std;
int main ()
{
        cout << "Hunaina Yasir" << "    " << "073";
        cout << "\n";

int i = 1;
  do
        {
    int j = 1;
    do {
      cout << j << " ";
      j++;
    } while (j <= i);

    cout << endl;
    i++;
  } while (i <= 10);

  return 0;
}
```

```cpp
#include <iostream>
using namespace std;
int main ()
{
    cout << "Hunaina Yasir" << "     " << "073";
    cout << "\n";

int i = 1;
    do
    {
        int j = 1;
        do {
            cout << j << " ";
            j++;
        } while (j <= i);

        cout << endl;
        i++;
    } while (i <= 10);

    return 0;
}
```

OUTPUT:

Hunaina Yasir   073
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4 5 6
1 2 3 4 5 6 7
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8 9
1 2 3 4 5 6 7 8 9 10

```
Hunaina Yasir    073
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4 5 6
1 2 3 4 5 6 7
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8 9
1 2 3 4 5 6 7 8 9 10
```

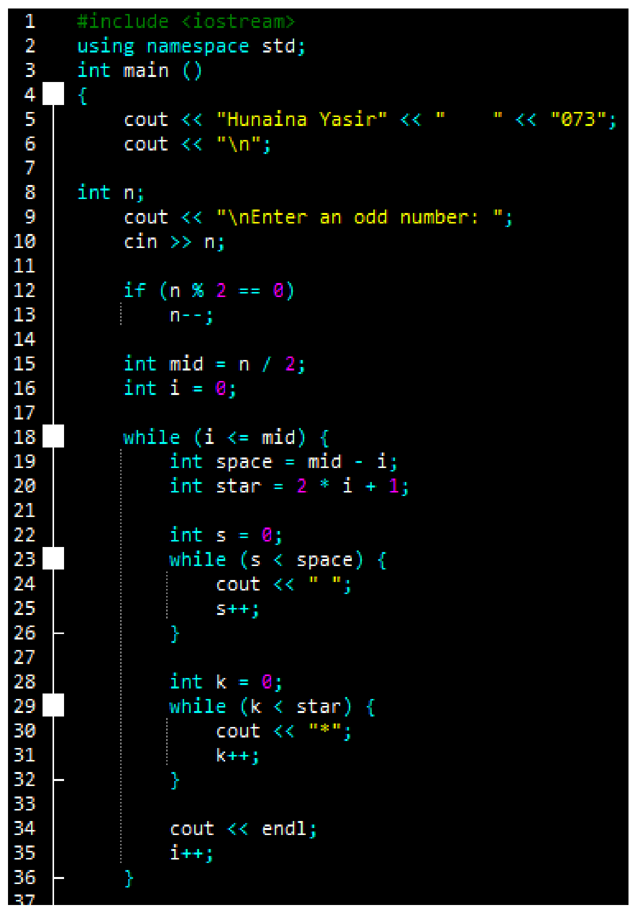
Exercises

6.1 Write a C++ program to read an odd integer from the user. If the user enters an even number, your program should decrement it by one to generate an odd number. Using this number as the number of rows, your program should then display a diamond of *s on the screen. Your program should have the following interface.

```
Enter an odd number: 5
  *
 ***
*****
 ***
  *
```

SOURCE CODE:

```
#include <iostream>
using namespace std;
int main ()
{
        cout << "Hunaina Yasir" << "    " << "073";
        cout << "\n";

int n;
  cout << "\nEnter an odd number: ";
  cin >> n;

  if (n % 2 == 0)
    n--;

  int mid = n / 2;
  int i = 0;

  while (i <= mid) {
    int space = mid - i;
    int star = 2 * i + 1;

    int s = 0;
    while (s < space) {
      cout << " ";
      s++;
    }

    int k = 0;
    while (k < star) {
      cout << "*";
      k++;
    }

    cout << endl;
    i++;
  }
```

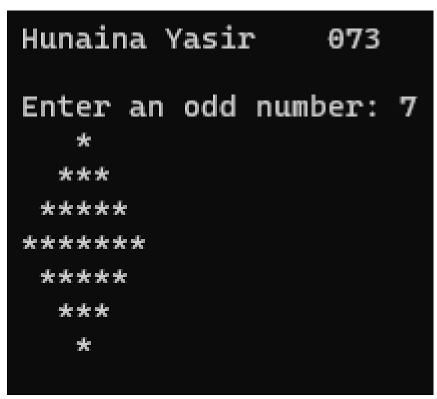


```
#include <iostream>
using namespace std;
int main ()
{
    cout << "Hunaina Yasir" << "    " << "073";
    cout << "\n";

int n;
    cout << "\nEnter an odd number: ";
    cin >> n;

    if (n % 2 == 0)
        n--;

    int mid = n / 2;
    int i = 0;

    while (i <= mid) {
        int space = mid - i;
        int star = 2 * i + 1;

        int s = 0;
        while (s < space) {
            cout << " ";
            s++;
        }

        int k = 0;
        while (k < star) {
            cout << "*";
            k++;
        }

        cout << endl;
        i++;
    }
```

```
  i = mid - 1;
  while (i >= 0) {
    int space = mid - i;
    int star = 2 * i + 1;

    int s = 0;
    while (s < space) {
      cout << " ";
      s++;
    }

    int k = 0;
    while (k < star) {
      cout << "*";
      k++;
    }

    cout << endl;
    i--;
  }

  return 0;
}
```

<u>OUTPUT:</u>

```
Hunaina Yasir    073

Enter an odd number: 7
   *
  ***
 *****
*******
 *****
  ***
   *
```



<u>6.2</u> Write a C++ program to produce the following multiplication table. You must use nesting of control structures.

1	2	3	4	5	6	7	8	9	10
2	4	6	8	10	12	14	16	18	20
3	6	9	12	15	18	21	24	27	30
4	8	12	16	20	24	28	32	36	40
5	10	15	20	25	30	35	40	45	50

6 12 18 24 30 36 42 48 54 60

7 14 21 28 35 42 49 56 63 70

8 16 24 32 40 48 56 64 72 80

9 18 27 36 45 54 63 72 81 90

10 20 30 40 50 60 70 80 90 100

SOURCE CODE:

```
#include <iostream>
using namespace std;
int main ()
{
            cout << "Hunaina Yasir" << "    " << "073";
            cout << "\n";

for (int i = 1; i <= 10; i++)
{
       for (int j = 1; j <= 10; j++)
                        {
          cout << i * j << " ";
       }
       cout << endl;
   }
   return 0;
}
```


```
#include <iostream>
using namespace std;
int main ()
{
      cout << "Hunaina Yasir" << "    " << "073";
      cout << "\n";

for (int i = 1; i <= 10; i++)
{
            for (int j = 1; j <= 10; j++)
            {
                  cout << i * j << " ";
            }
            cout << endl;
      }
      return 0;
}
```


OUTPUT:

```
0983
0984
0985
0986
0987
0988
0989
0990
0991
0992
0993
0994
0995
0996
0997
0998
0999
```

```
9981
9982
9983
9984
9985
9986
9987
9988
9989
9990
9991
9992
9993
9994
9995
9996
9997
9998
9999
```

It prints from numbers 0983 to 9999.

<u>6.3</u> Write a C++ program which reads in a number from user, stores it in a variable named ‘N’ and calculates the sum of powers using the following formula.

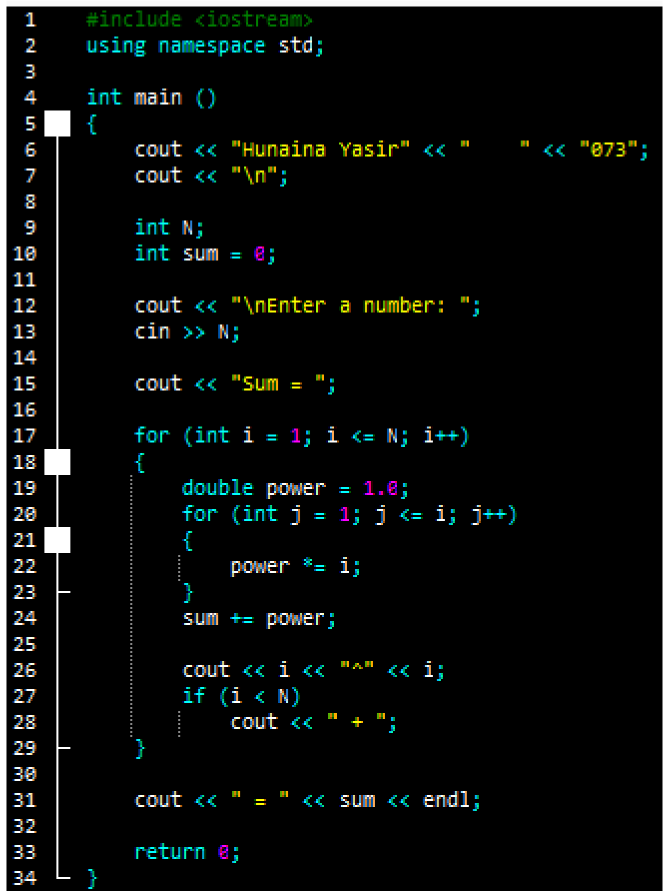
Sum = $1^1 + 2^2 + 3^3 + \ldots\ldots + N^N$

Your program should have the following interface.

```
Enter a number: 4
Sum = 1^1 + 2^2 + 3^3 + 4^4 = 288
```

<u>SOURCE CODE:</u>

```cpp
#include <iostream>
using namespace std;

int main ()
{
        cout << "Hunaina Yasir" << "    " << "073";
  cout << "\n";

  int N;
  int sum = 0;

  cout << "\nEnter a number: ";
  cin >> N;

  cout << "Sum = ";

  for (int i = 1; i <= N; i++)
  {
     double power = 1.0;
     for (int j = 1; j <= i; j++)
     {
        power *= i;
     }
     sum += power;

     cout << i << "^" << i;
     if (i < N)
        cout << " + ";
  }

  cout << " = " << sum << endl;

  return 0;
}
```




```cpp
#include <iostream>
using namespace std;

int main ()
{
    cout << "Hunaina Yasir" << "    " << "073";
    cout << "\n";

    int N;
    int sum = 0;

    cout << "\nEnter a number: ";
    cin >> N;

    cout << "Sum = ";

    for (int i = 1; i <= N; i++)
    {
        double power = 1.0;
        for (int j = 1; j <= i; j++)
        {
            power *= i;
        }
        sum += power;

        cout << i << "^" << i;
        if (i < N)
            cout << " + ";
    }

    cout << " = " << sum << endl;

    return 0;
}
```


<u>OUTPUT:</u>

Hunaina Yasir   073
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4 5 6
1 2 3 4 5 6 7
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8 9
1 2 3 4 5 6 7 8 9 10